



REAL-TIME AI-IOT ELEPHANT DETECTION AND ACOUSTIC DETERRENT FOR SRI LANKAN RAILWAYS

Project Id - 26-25j-331

PP2 Comments

Topic	Real-Time AI-IoT Elephant Detection and Acoustic Deterrent for Sri Lankan Railways			
Project ID	25-26J-331			
Specialization	IT			
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PanelName				
Detailed Comments	Good progress. Addressed all comments given in PP2	Good progress. Addressed all comments given in PP2	Good progress. Addressed all comments given in PP2	Good progress. Addressed all comments given in PP2



Problem in Brief

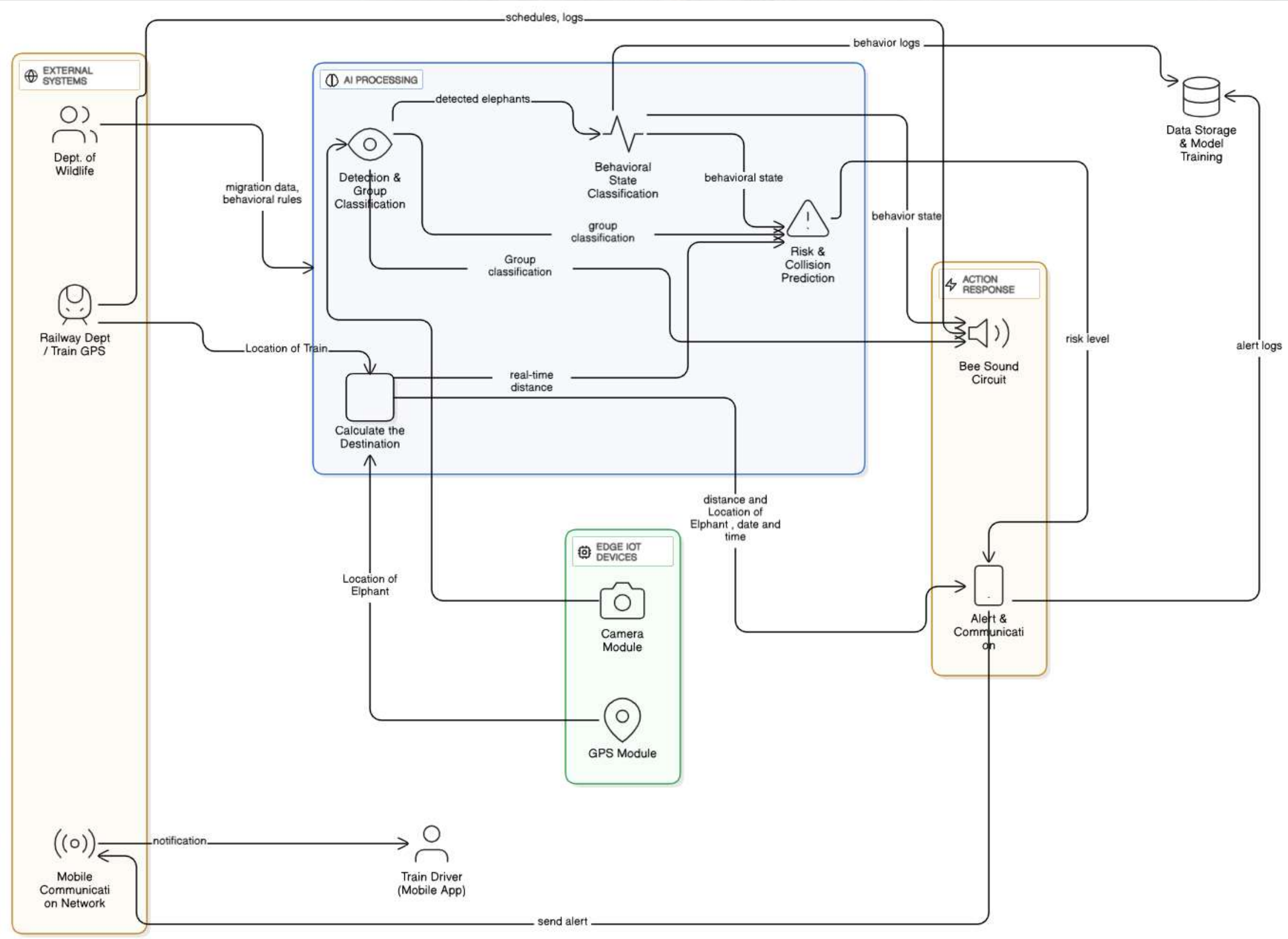
- Frequent train–elephant collisions in railway tracks, causing loss of wildlife
- Lack of real-time detection and early warning systems for train drivers
- High risk to passenger safety and railway infrastructure
- Existing solutions are manual, inefficient, and unable to predict elephant behavior

Solution Overview

- Provides a fully automated real-time detection system using YOLOv8
- Integrates behavior analysis to identify elephant actions , which is not available in existing systems
- Uses accurate distance calculation for precise risk evaluation
- Implements LoRa communication for long-range, reliable, and low-power alert transmission
- Combines AI, IoT, and mobile application into a single integrated, scalable, and cost-effective solution



SYSTEM OVERVIEW DIAGRAM



Real-Time Elephant Detection, Counting and Group Classification

Key Features

- Real-time elephant detection using deep learning
- Accurate counting of multiple elephants in a scene
- Classification of elephants into Adult and Calf
- Group type identification (Single / Herd / Family group)
- Edge deployment on Raspberry Pi for real-time processing

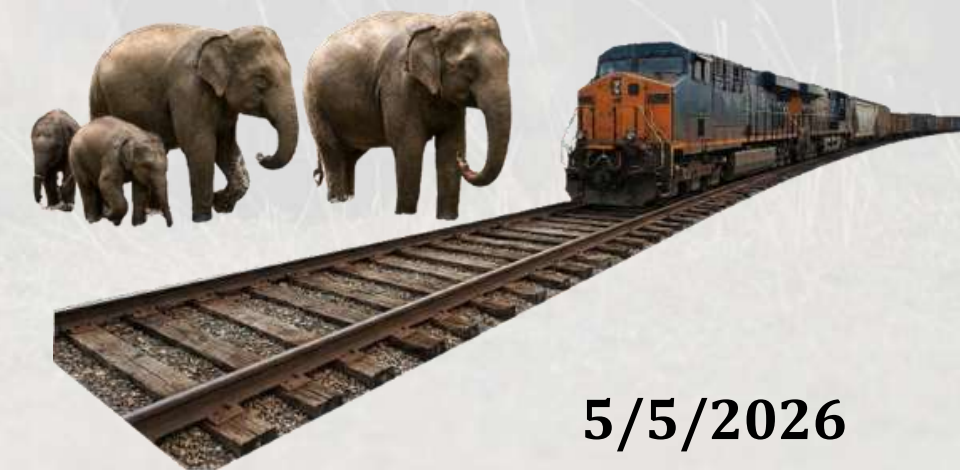
Technologies Used:

- YOLOv8: Deep learning model for fast and accurate object detection
- Python: Used for model development, training, and system integration
- OpenCV: Image processing and real-time frame handling
- Raspberry Pi: Edge device for running the model in real-time
- Camera Module: Captures live video



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Behavioral State Classification

Key Features :

- Classifies elephant behavior as calm or aggressive using CNN
- Analyzes movement patterns such as speed and direction
- Enhances risk prediction accuracy using behavior data
- Processes real-time image frames for continuous monitoring

Technologies Used:

- Convolutional Neural Networks (CNN): For behavior classification
- TensorFlow / Keras: Model training and implementation
- Python: Development and integration
- OpenCV: Image and motion analysis



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Distance Calculation & Risk Prediction

Key Features :

- Calculates real-time distance between the train and the pillar which elephant are detected using GPS data
- Uses Haversine formula for accurate geographic distance measurement
- Transmits alerts to the train driver using LoRa long-range communication
- Generates risk level based on distance, group type, and behavior inputs

Technologies

- ~~Used:~~ Mobile GPS – Provides real-time train location
- Haversine Formula – Calculates distance using latitude and longitude
- ESP32 – Controls LoRa communication and data transmission
- LoRa Module – Sends long-range alerts to the train driver
- Mobile Application – Displays distance and risk level



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Mobile Application & IoT System

Key Features :

- Provides real-time alerts to train drivers through a mobile application
- Displays risk level, distance, and location information clearly
- Uses an IoT-based sound device to safely drive elephants away from tracks
- Implements adaptive alert system where sound intensity changes based on risk level

Technologies Used:

- Mobile App Development: User interface for alerts
- IoT Device (Buzzer / Speaker): Generates sound warnings
- Microcontroller (ESP32): Controls IoT device operations



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Results

- Achieved accurate real-time elephant detection using YOLOv8
- Successful behavior classification (calm vs aggressive) using CNN
- Reliable distance calculation and risk level generation
- Effective LoRa communication and alert delivery system

User Acceptance / Feedback

- System provides clear and timely alerts to users
- Easy to understand risk levels and notifications
- Helps in early decision-making for train drivers
- Considered useful and practical for real-world application



Limitations

- Detection accuracy depends on training data, which may be limited in real-world scenarios
- Hardware limitations of Raspberry Pi can impact processing speed and performance
- Dependence on GPS and communication networks may affect reliability in remote areas

Future Improvement

- Enhance detection using advanced sensors (thermal imaging, night vision) for better performance in all conditions
- Improve AI models with larger datasets to increase accuracy and reduce false detections
- Implement automated train control integration (e.g., speed reduction or braking systems)
- Develop a fully scalable system for deployment across national railway networks





Q & A



